



Introduction to ZKsync

A learning group for ZK and SNARK application development

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Respectful collaboration starts here

**ALL ARE
WELCOME**

**TODO
BIENVENIDOS**

**みなさん
歓迎します。**

www.lfdecentralizedtrust.org/code-of-conduct

Agenda



- Logistics
- Zero knowledge proofs - summary
- ZK Rollups
- ZK Rollup high level overview
- ZKsync
- ZKsync - components
- ZKsync - features and chains
- Demo, hello world
- Challenge
- Links, Resources, Literature
- Q&A

Logistics

Every month or two months

Similar structure, similar logistics + **guest lectures**

Theory:

- Look tables,
- Zk variations: STARKs, FRI
- recursive SNARKs

Programming / platforms:

- ZoKrates, Leo
- Zinc / ZKSync

Applications:

- ZK bridges
- ZK Virtual Machines
- ZK Identity / verifiable credential - presentations
- Voting



Discord channel:

<https://discord.com/channels/905194001349627914/1329201532628898036>

Meetup.com:

<https://www.meetup.com/lfdt-hungary/events/305634614/>

Repo with all the contents:

<https://github.com/LF-Decentralized-Trust-labs/zk-learning-group>

Zero knowledge proofs - summary

Verifiable computation with cryptographic proof

Scarce resource:

- limited verification time: no re-execution
- storage limits: smaller proof
- limited information

Computation: arithmetic circuit : $C(x, w) \rightarrow F$

- x public input
- w private input, witness
- high level computation
- arithmetic circuit
- polynomials

Prover algorithm: $P(pp, x, w) \rightarrow \text{proof}$

Verifier algorithm: $V(vp, x, \text{proof}) \rightarrow \text{accept / reject}$

Zero Knowledge Proof



Interactive / non-interactive proofs

ZK Rollups

Transactions are executed off-chain

Rollup, sequencer

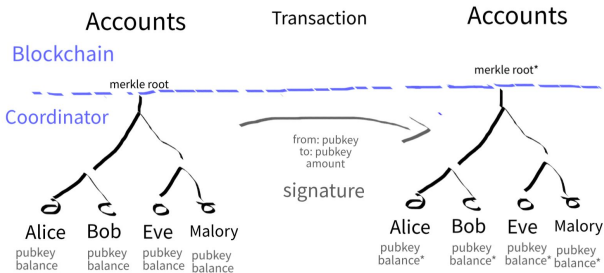
Execution result is exported back to the blockchain

Execution result is hardened by a zero knowledge proof

Zero knowledge proof if validated on-chain (smart contract)

Result is valid only if the ZK proof on the correct execution is valid as well.

E.g. Arbitrum



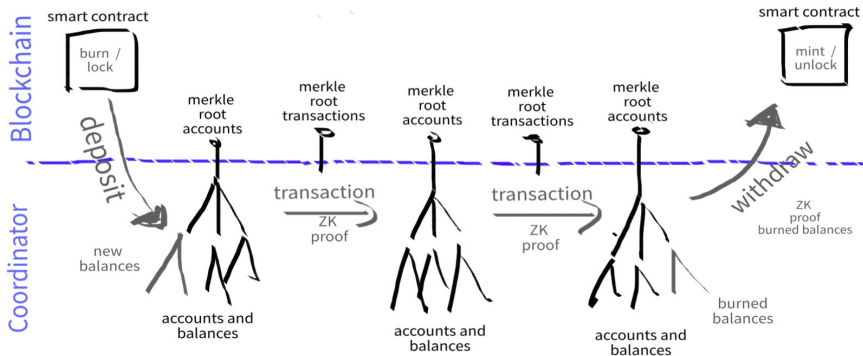
ZK Rollup high level overview

Off-chain
transactions

Payment / token
transfer

Transactions are
executed in batch /
block by a
coordinator

Deposit and
withdraw
transactions



ZKsync

Layer 2 scaling protocol for Ethereum

Zero Knowledge rollup, zkEVM

EVM compatible off-chain smart contract execution

processing transactions off-chain and verify them on Ethereum

Creating interoperable ZK-powered blockchains

ZKsync chains share users and liquidity

ZKEra: general purpose ZK rollup

ZK Stack: open-source framework for L2, L3 rollups



<https://www.zksync.io/>

ZKsync - components

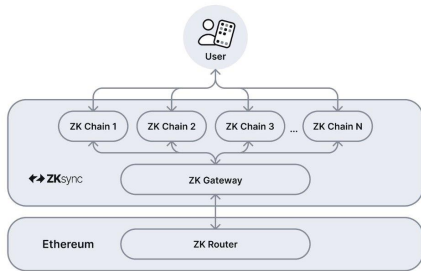
Core Network Components

Sequencer: Orders, executes transactions off-chain, and produces initial state updates.

EraVM: A custom virtual machine optimized to execute smart contract bytecode while enabling efficient zero-knowledge proof generation.

Prover (Airbender / Boojum): A scalable cryptographic proof generator that computes validity proofs for transaction batches.

Verifier Smart Contract: An L1 contract deployed on Ethereum that cryptographically verifies zero-knowledge proofs before finalizing state transitions.



<https://www.zksync.io/>

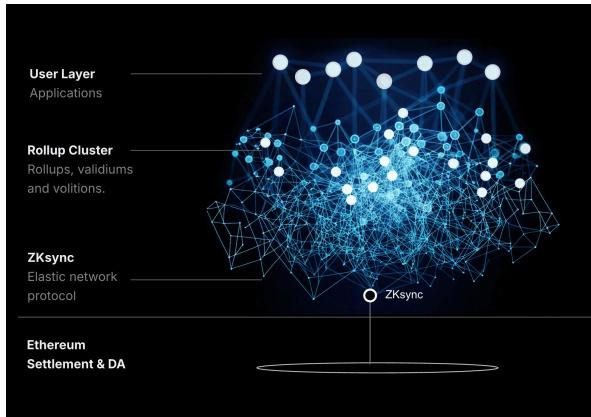
ZKsync - components

ZK Stack & OS Components

ZKsync OS: The modular execution framework featuring a Bootloader (initializes the system), Execution Environments (interpreters for bytecode), and a System Layer (manages low-level memory and input/output)

ZKsync OS Server: The high-performance server architecture containing the sequencer, Web3 JSON-RPC API, and batcher.

ZK Chains & Gateway: Independent custom rollups built on the ZK Stack that connect via middleware gateways and routers for shared liquidity and trustless interoperability



<https://docs.zksync.io/zk-stack/zk-chains>

ZKsync - features and chains



Interoperability

- native, bridge-less cross-chain communication system
- inerop communication system (messages, calls)
- Elastic chain: ecosystem of interconnected ZK chains, through native protocol-level bridges:
 - Native interop
 - Shared liquidity
 - Unified UX

Validium: L2 rollup storing data on external storage

Core Network

- zkSync Mainnet (Era)

Elastic Chain Networks

- GRVT Mainnet
- Cronos zkEVM Mainne
- ...

Ecosystem / App Chains

- Abstract Chain
- ZERO Network

Institutional applications

- Prividium

Demo



**Hello word solidity
contract deployed to
ZKsync**

Challenge



**Build a simple (9,9)
Sudoku in Solidity and
deploy it to a ZKsync
chain or rollup**

Links, Resources, Literature



ZK sync

<https://www.zksync.io/>

A complete guide to zkSync

<https://support.token.im/hc/en-us/articles/4405480116889-A-complete-guide-to-zkSync>

zkSync Guide - The future of Ethereum scaling

<https://soliditydeveloper.com/zkSync>

ZKsync web3 examples

<https://codesandbox.io/examples/package/zksync-web3>

CryptoZombies zkSync

<https://cryptozombies.io/en/course/>



Happy Hunting for the SNARK :)

Q & A

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